

LAB 09

Task 01:

```
#include <iostream>
using namespace std;
class Transaction {
protected:
    int accountNumber;
    double amount;
public:
    Transaction(int accNum, double amt) : accountNumber(accNum), amount(amt) {}
    virtual void processTransaction() = 0;
};
class Deposit : public Transaction {
public:
    Deposit(int accNum, double amt) : Transaction(accNum, amt) {}
    void processTransaction() override {
        cout << " Deposit Transaction" << endl;
        cout << "Account: " << accountNumber << "\nAmount: " << amount << endl;
        cout << "Status: Amount Deposited Successfully" << endl << endl;
    }
};
class Withdrawal : public Transaction {
public:
    Withdrawal(int accNum, double amt) : Transaction(accNum, amt) {}
    void processTransaction() override {
        cout << "Withdrawal Transaction" << endl;
        cout << "Account: " << accountNumber << "\nAmount: " << amount << endl;
        cout << "Status: Amount Withdrawn Successfully" << endl << endl;
    }
};
class Transfer : public Transaction {
private:
    int targetAccountNumber;
public:
    Transfer(int accNum, double amt, int targetAcc) : Transaction(accNum, amt), targetAccountNumber(targetAcc) {}
    void processTransaction() override {
        cout << "Transfer Transaction" << endl;
        cout << "From Account: " << accountNumber << "\nTo Account: " << targetAccountNumber << "\nAmount: " << amount << endl;
        cout << "Status: Transfer Completed Successfully" << endl << endl;
    }
};
int main() {
    Deposit acc(1001, 5000);
    acc.processTransaction();
    return 0;
}
```

Output:

▼ TERMINAL

```
cd "/home/rumaisa-abbasi/Desktop/Lab 09/" && g++ task1.cpp -o task1 && "/home/rumaisa-abbasi/Desktop/Lab 09/"task1
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 09$ cd "/home/rumaisa-abbasi/Desktop/Lab 09/" && g++ task1.c
Deposit Transaction
Account: 1001
Amount: 5000
Status: Amount Deposited Successfully
```

Task 02:

```
using namespace std;
class Notification {
protected:
    string recipient;
public:
    Notification(string res) : recipient(res) {}
    virtual void sendNotification(string message) = 0;
    virtual ~Notification() {}
};

class SMSNotification : public Notification {
public:
    SMSNotification(string res) : Notification(res) {}
    void sendNotification(string message) override {
        cout << "SMS Notification" << endl;
        cout << "To: " << recipient << endl;
        cout << "Message: " << message << endl;
        cout << "Status: Sent Successfully" << endl;
    }
};

class EmailNotification : public Notification {
public:
    EmailNotification(string res) : Notification(res) {}
    void sendNotification(string message) override {
        cout << "\nEmail Notification" << endl;
        cout << "To: " << recipient << endl;
        cout << "Message: " << message << endl;
        cout << "Status: Sent Successfully" << endl;
    }
};

class PushNotification : public Notification {
public:
    PushNotification(string res) : Notification(res) {}
    void sendNotification(string message) override {
        cout << "\nPush Notification" << endl;
        cout << "To: " << recipient << endl;
        cout << "Message: " << message << endl;
        cout << "Status: Sent Successfully" << endl;
    }
};

int main() {
    SMSNotification sms("03001234567");
    sms.sendNotification("Your OTP is 1234");
    return 0;
}
```

Output:

```
cd "/home/rumaisa-abbasi/Desktop/Lab 09/" && g++ task2.cpp -o task2 && ./task2
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 09$ cd "/home/rumaisa-abbasi/Desktop/Lab 09/"
SMS Notification
To: 03001234567
Message: Your OTP is 1234
Status: Sent Successfully
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 09$
```